NUMBER THEORY

MODULE 3

The Theory of Congruences

SUBMODULE 1

Introduction to Congruences

OBJECTIVES

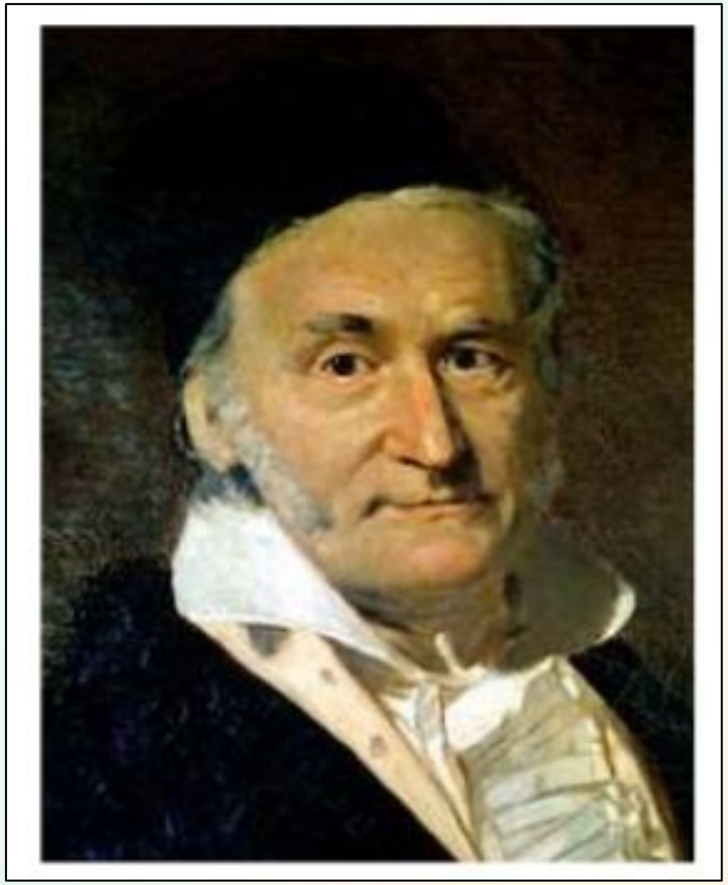
- Define congruences, residue classes and least residues.
- Demonstrate understanding of the basic concepts of modular arithmetic.

Introduction to Congruences

Theory of Congruences

- Also known as arithmetic of remainders
- An approach to divisibility questions
- Introduced by Carl Friedrich Gauss in 1801
- Gauss adopted the symbol '≡' for congruence
- According to Gauss:

"If a number *n* measures the difference between two numbers *a* and *b*, then *a* and *b* are said to be congruent with respect to *n*; if not, incongruent."



Introduction to Congruences

Definition: Let *m* be a fixed positive integer. Two integers *a* and *b* are said to be *congruent modulo m* symbolized by

$$a \equiv b \bmod m$$

if $m \mid (a - b)$ provided that $a - b = km$ for some integer k.

Examples:

$3 \equiv 24 \pmod 7$ $\quad -31 \equiv 11 \pmod 7$ $\quad -15 \equiv -64 \pmod 7$

$13 \not\equiv 5 \pmod 9$ since $9 \nmid (13 - 5) = 8$

- Also interpreted as "*a* and *b* have the same *'remainder'* when they are divided by *m*."
- In working with congruences, sometimes there is a need to translate them into equality.

Theorem If a and b are integers, then $a \equiv b \pmod m$ if and only if there is an integer k such that $a = b + km$.

Introduction to Congruences

Properties of Congruences

Let m be a positive integer and let a, b, c, d be integers. Then

1. $a \equiv a \bmod m$ → **Reflexive Property**
2. If $a \equiv b \bmod m$, then $b \equiv a \bmod m$. → **Symmetric Property**
3. If $a \equiv b \bmod m$ and $b \equiv c \bmod m$, then $a \equiv c \bmod m$. → **Transitive Property**
4. (a) If $a \equiv qm + r \bmod m$, then $a \equiv r \bmod m$.

   (b) Every integer a is congruent mod m to exactly one of $0, 1, \ldots, m-1$.

Introduction to Congruences

Properties of Congruences

C_1: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a \pm c \equiv b \pm d \pmod{m}$.

C_2: If $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{m}$.

C_3: If $a \equiv b \pmod{m}$ and $c \geq 1$, then $ac \equiv bc \pmod{mc}$.

C_4: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$.

C_5: If $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$ for all $n \geq 1$.

C_6: $a \equiv 0 \pmod{m}$ if and only if $m|a$.

C_7: If $ac \equiv bc \pmod{m}$ and $(c, m) = 1$, then $a \equiv b \pmod{m}$.

C_8: If $ac \equiv bc \pmod{m}$ and $(c, m) = n$, then $a \equiv b \pmod{m/n}$.

C_9: If $a \equiv b \pmod{m}$ and $n|m$, then $a \equiv b \pmod{n}$.

Introduction to Congruences

Reduced Modulo

- The resulting congruence when a common factor is cancelled from a congruence.

Example:

$$110 \equiv 50 \pmod{15}$$

That is,

$$10 \cdot 11 \equiv 10 \cdot 5 \pmod{15}$$

Now $(10, 15) = 5$. Therefore, if we cancel 10 from both sides of the last congruence, we get

$$11 \equiv 5 \pmod{15/5}$$

That is,

$$11 \equiv 5 \pmod{3}$$

Applying Property C8.

Introduction to Congruences

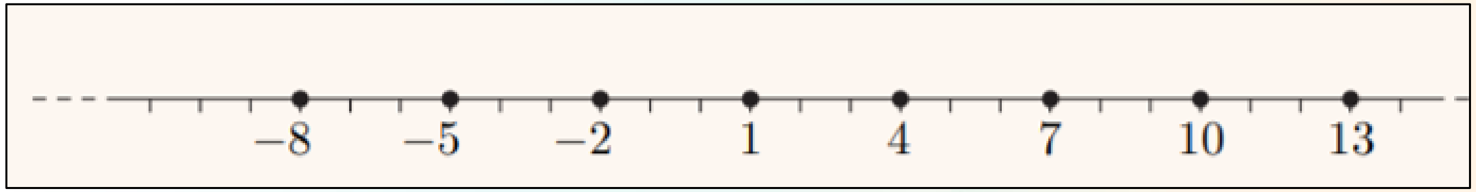
Congruence/Residue Class

Given any integer a, the collection of all integers congruent to a modulo n is known as the **residue class**, or **congruence class**, of a modulo n.

- The word "**residue**" means "**remainder**".
- This term is used because the residue class of *a modulo n* is the class of those integers that have the same remainder on division by *n* as *a* does.

Examples:

The residue class of 1 mod 3
The residue class of 4 mod 3
The residue class of -2 mod 3



Introduction to Congruences

Least Residues

The **least residue** of a modulo n is the remainder r that you obtain when you divide a by n.

The integer r is one of the numbers $0, 1, \ldots, n-1$, and it satisfies

$$a \equiv r \pmod{n}.$$

Examples:

	Proof	Least Residue
35 (mod 8)	35 = 4(8) + 3	3
35 (mod 7)	35 = 5(7) + 0	0
-13 (mod 10)	-13 = (-2)(10) + 7	7

The set of least positive residues (mod *m*) is the set {0, 1, 2, . . ., m – 1}

Example: Set of least positive residues (mod 4) → {0, 1, 2, 3}
Set of least positive residues (mod 7) → {0, 1, 2, 3, 4, 5, 6}

Introduction to Congruences

Modular Arithmetic (Addition and Subtraction)

- The application of the usual arithmetic operations for congruences.

Addition and subtraction rules for congruences

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d \pmod{n}$$
$$a - c \equiv b - d \pmod{n}.$$

- Addition or subtraction to both sides of a congruence preserves the congruence.

Examples:

$19 \equiv 3 \pmod{8}$

$19 + 7 \equiv 3 + 7 \pmod{8} \rightarrow 26 \equiv 10 \pmod{8}$

$19 - 4 \equiv 3 - 4 \pmod{8} \rightarrow 15 \equiv -1 \pmod{8}$

Find the least residue of $171 + 169 \pmod{17}$

Solution:

$171 + 169 = 340 \equiv \mathbf{0} \pmod{17}$

Introduction to Congruences

Modular Arithmetic (Multiplication)

Multiplication rule for congruences

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$ac \equiv bd \pmod{n}.$$

- Multiplication to both sides of a congruence preserves the congruence.
- Simpler because integers are often replaced with simpler numbers before multiplying them.

Examples:

19 ≡ 3 (mod 8)

(19)(2) ≡ (3)(2) (mod 8) ➔ 38 ≡ 6 (mod 8)

Find the least residue of 52 x 37 (mod 7)

Solution:

52 ≡ 3 (mod 7) and 37 ≡ 2 (mod 7)

52 x 37 ≡ 3 x 2 ≡ **6** (mod 7)

Introduction to Congruences

Modular Arithmetic (Division)

- It is not necessarily true that congruence is preserved when both sides of the congruence are divided by an integer.
- In order to preserve the congruence, the GCD of the integer to be used as divisor and the modulo *m* must be equal to 1.

Examples:

$30 \equiv 42 \pmod 4$

Using **3** as divisor: GCD(3, 4) = 1

$30/3 \equiv 42/3 \pmod 4$
$10 \equiv 14 \pmod 4$

$30 \equiv 42 \pmod 4$

Using **6** as divisor: GCD(6, 4) = 2

$30/6 \equiv 42/6 \pmod 4$
$5 \equiv 7 \pmod 4$
*** not congruent**

Introduction to Congruences

Modular Exponentiation

- Used to find the least residue of congruences involving large powers of integers.
- Such congruences are of the form **a^k (mod m).**

Power rule for congruences

If $a \equiv b \pmod{n}$, and m is a positive integer, then

$$a^m \equiv b^m \pmod{n}.$$

Example:

Finding the least residue of **19^5 (mod 9).**

Since:

$$19 \equiv 1 \text{ (mod 9)}$$

Then:

$$19^5 \equiv 1^5 \equiv 1 \text{ (mod 5)}$$

Introduction to Congruences

Modular Exponentiation – Finding Least Residue

Strategy 1: **Replacing the integer by its least positive residue.**

Note: Applicable whenever a number greater than or equal to a modulus is generated.

Example: $233^5 \pmod{11}$

Since $233 > 11 \Rightarrow 233 = 21(11) + 2 \Rightarrow 233 \equiv 2 \pmod{11}$

$233^5 \equiv 2^5 = 32 \equiv 10 \pmod{11}$

Introduction to Congruences

Modular Exponentiation – Finding Least Residue

Strategy 2: **Making use of the binary representation of the exponent k.**

Note 1: This strategy is the general procedure for modular exponentiation.

Note 2: Any even power of a given base can be computed by squaring, since:

$$a^{2k} = (a^k)^2$$

Note 3: Any odd power of a given base can be computed by multiplying the base by a square, since:

$$a^{2k+1} = a(a^{2k})$$

Steps: 1. Express the exponent ***k*** in binary notation.

2. Find the least positive residues of a, a^2, a^4, . . ., a^{2^k} (mod m), by successively squaring and reducing modulo m.

3. Compute a^k by multiplying the least positive residues of the appropriate powers of 2.

Introduction to Congruences

Modular Exponentiation – Finding Least Residue

Example: **$15^{10} \pmod{31}$**

1. $(10)_{10} = (1010)_2$ ➔ Powers of 2 are 1, 2, 4, 8

2. $15^1 = 15 \pmod{31}$
   $15^2 = (15^1)^2 = 225 \equiv 8 \pmod{31}$
   $15^4 = (15^2)^2 \equiv 8^2 = 64 \equiv 2 \pmod{31}$
   $15^8 = (15^4)^2 \equiv 2^2 = 4 \equiv 4 \pmod{31}$

3. $15^{10} = 15^{8+2} = (15^8)(15^2) \equiv (4)(2) = 32 \equiv 1 \pmod{31}$

Introduction to Congruences

Modular Exponentiation – Finding Least Residue

Strategy 3: Making use of squaring operations

Note: The number of squaring operations needed to compute a^k is **[log$_2$ k]**.

Example: $7^{19} \pmod{31}$

Step 1:

$$\begin{aligned}
19 &= 2(9) + 1 = 18 + 1 \\
18 &= 2(9) \\
9 &= 2(4) + 1 = 8 + 1 \\
8 &= 2(4) \\
4 &= 2(2) \\
2 &= 2(1)
\end{aligned}$$

compute only 7^2, 7^4, 7^8, 7^9, 7^{18}, and $7^{19} \pmod{31}$:

$$\begin{aligned}
7^2 &= (7^1)^2 = 49 \equiv 18 \pmod{31} \\
7^4 &= (7^2)^2 \equiv 18^2 = 324 \equiv 14 \pmod{31} \\
7^8 &= (7^4)^2 \equiv 14^2 = 196 \equiv 10 \pmod{31} \\
7^9 &= 7(7^8) \equiv 7(10) \equiv 70 \equiv 8 \pmod{31} \\
7^{18} &= (7^9)^2 \equiv 8^2 = 64 \equiv 2 \pmod{31} \\
7^{19} &= 7(7^{18}) \equiv 7(2) = 14 \pmod{31}
\end{aligned}$$

The number of squaring will be **[log$_2$ 19] = 4**.

Introduction to Congruences

Activity 1

1. Show that $13 \equiv 1 \pmod{2}$ holds
2. Reduce $650 \equiv 350 \pmod{75}$ as much as possible by cancellation
3. Plot the residue class of $0 \pmod{5}$ on a number line
4. Find the least residue of $17 \pmod{10}$
5. Find the least residue of $7 + 3 \pmod{6}$
6. Find the least residue of $3 \times 6 \pmod{7}$
7. Find the least residue of $2^{200} \pmod{47}$

Formative Assessment 8

Check Canvas

SUBMODULE 2

Linear Congruences

OBJECTIVES

- Apply divisibility tests to large numbers.
- Find solutions for a linear congruences.

Linear Congruences

Special Divisibility Tests

Divisibility by Powers of 2

An integer is divisible by $\mathbf{2^j}$ if and only if its last ***j*** digits is divisible by $\mathbf{2^j}$.

Example:

32688048

➔ divisible by 2^1 because $2^1 = 2$ and $2 \mid 8$
➔ divisible by 2^2 because $2^2 = 4$ and $4 \mid 48$
➔ divisible by 2^3 because $2^3 = 8$ and $8 \mid 048$
➔ divisible by 2^4 because $2^4 = 16$ and $16 \mid 8048$
➔ not divisible by 2^5 because 32 does not divide 88048

Linear Congruences

Special Divisibility Tests

Divisibility by Powers of 5

An integer is divisible by $\mathbf{5^j}$ if and only if its last $\boldsymbol{j}$ digits is divisible by $\mathbf{5^j}$.

Example:

15535375

➔ divisible by 5^1 because $5^1 = 5$ and $5 \mid 5$
➔ divisible by 5^2 because $5^2 = 25$ and $25 \mid 75$
➔ divisible by 5^3 because $5^3 = 125$ and $125 \mid 375$
➔ not divisible by $5^5 = 625$ because 625 does not divide 5375

Linear Congruences

Special Divisibility Tests

Divisibility by 3 and 9

An integer is divisible by **3 or 9** if and only if the sum of its digits is divisible by **3**, or by **9**, respectively.

Example:

4127835 ➔ 4 + 1 + 2 + 7 + 8 +3 + 5 = 30 → 3 | 30 but 9 does not divide 30
Thus, **4127835 | 3**

8675038695 ➔ 8 + 6 + 7 + 5 + 0 + 3 + 8 + 6 + 9 + 5 = 57 ➔ 5 + 7 = 12
➔ 12 | 3. Thus, **8675038695 | 3**

5841 ➔ 5 + 8 + 4 + 1 = 18 → 18 | 9. Thus, **5841 | 9**

Linear Congruences

Special Divisibility Tests

Divisibility by 11

An integer is divisible by 11 if and only if the alternating sum of its digits is divisible by 11.

Examples:

a) 1571724 ➔ $4 - 2 + 7 - 1 + 7 - 5 + 1 = 11$
➔ divisible by 11. Thus, **1571724 | 11**

b) 673148 ➔ $8 - 4 + 1 - 3 + 7 - 6 = 3$
➔ not divisible by 11. Thus **673148 does not divide 11**

Linear Congruences

Special Divisibility Tests

Divisibility by 7, 11 and 13

An integer is divisible by 7, 11 or 13 if and only if the alternating sum and difference of blocks of three digits is divisible by 7, 11 or 13.

Examples:

a) 59358208 ➔ 208 – 358 + 59 = –91

➔ divisible by 7 and 13, but not by 11. Thus, **59358208 | 7** and **59358208 | 13**

Linear Congruences

Definition:

A **linear congruence** is a congruence of the form

$$ax \equiv b \pmod{n},$$

where a and b are known, and x is unknown.

Examples:

$9x \equiv 12 \pmod{15}$

$3x \equiv 1 \pmod{15}$

Linear Congruences

- Any number ***x*** for which the linear congruence is true is a *solution* of the linear congruence.
- If ***x*** is a solution of a linear congruence, then any number in the same residue class as *x modulo b* is also a solution.

For instance, since 3 is a solution of the linear congruence $3x \equiv 4 \pmod{5}$, every member of the congruence class $[3] = \{ \ldots, -7, -2, 3, 8, 13, \ldots \}$ is also a solution.

Linear Congruences

The linear congruence $ax \equiv b \pmod{m}$ is solvable if and only if $d \mid b$, where $d = \mathrm{GCD}(a, m)$. If $d \mid b$, then it has d incongruent solutions.

Examples:

14x ≡ 12 (mod 18)

a = 14, b = 12, m = 18

d = GCD(14, 18) = 2
2 | 18 … Solution exists
2 solutions

10x ≡ 2 (mod 5)

a = 10, b = 2, m = 5

d = GCD(10, 5) = 5
5 does not divide 2 … No solution.

A linear congruence where the multiplier ***a*** and the modulus ***m***, where $m > 0$, are relatively prime has a unique solution modulo m.

Example:

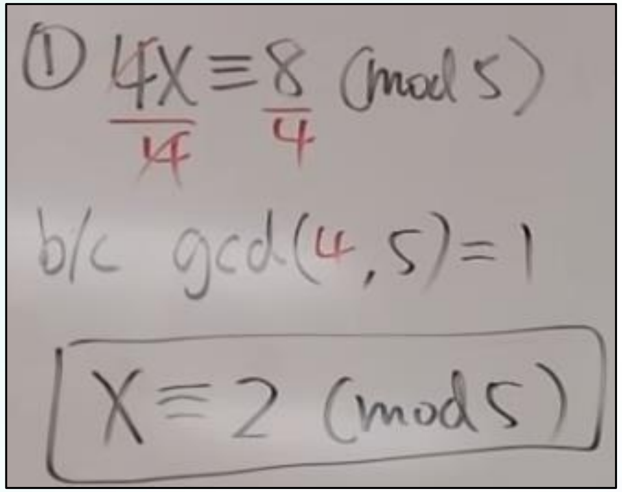
4x ≡ 8 (mod 5)

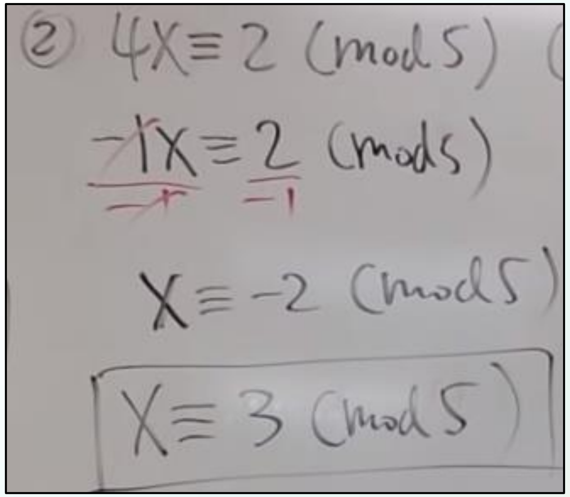
a = 4, m = 5

Linear Congruences

In general, the solution of a linear congruence $ax \equiv b \pmod{m}$ if it exists, will have the form $x \equiv t \pmod{m}$.

Examples:





Linear Congruences

The linear congruence $\boldsymbol{ax \equiv b \pmod{m}}$ is equivalent to the LDE $\boldsymbol{ax - my = b}$.
So, the congruence is solvable if and only if the LDE is solvable. As mentioned in the previous lessons, $ax \equiv b \pmod{m}$ is solvable if and only if $d \mid b$.

If $ax \equiv b \pmod{m}$ is solvable and x_0, y_0 is one specific solution, then any other solution has the following form for some choice of $\boldsymbol{t}$.

$$x = x_0 + \left(\frac{m}{d}\right)t \quad and \quad y = y_0 + \left(\frac{a}{d}\right)t$$

where $x = x_0$ and $y = y_0$ is a particular solution of the equation. The values of x given above,

$$x = x_0 + \left(\frac{m}{d}\right)t$$

are the solutions of the linear congruence.

Linear Congruences

The complete set of incongruent solutions is obtained by taking:

$$x = x_0 + \left(\frac{m}{d}\right)t$$

where *t* ranges through a complete set of residues modulo *d*.
That is:

$$t = 0, 1, 2, \ldots, d - 1.$$

Linear Congruences

Example :

$9x \equiv 12 \pmod{15}$

a = 9, b = 12, m = 15

d = GCD(9, 15) = 3

3 | 12 … Solution exists

3 solutions

$9x - 15y = 12$

By Euclid's Algorithm:

15 = (1)(9) + 6

9 = (1)(6) + 3

6 = (2)(3) + 0

Making each line as subject of the remainder:

6 = (1)(15) – (1)(9)

3 = (1)(9) – (1)(6)

By backward substitution:

3 = (1)(9) – (1) **[(1)(15) – (1)(9)]**

3 = (1)(9) – (1)(15) + (1)(9)

[3 = (2)(9) – (1)(15)] x 4

12 = (8)(9) – (4)(15)

t = 0, 1, 2 → complete set of residues for d = 3

$$x = x_0 + \left(\frac{m}{d}\right) t$$

$x \equiv 8 \pmod{15}$,

$x \equiv 13 \pmod{15}$,

$x \equiv 18 \equiv 3 \pmod{15}$,

Linear Congruences

Modular Inverses

- Useful in solving linear congruences.
- The linear congruence $ax \equiv 1 \pmod{m}$ has a unique solution if and only if $(a, m) = 1$.
- This means that when $(a, m) = 1$, there is unique least residue x such that $ax \equiv 1 (\text{mod } m)$.
- In $ax \equiv 1(\text{mod } m)$, a is said to be invertible and x is called an **inverse of *a* (mod *m*)**, denoted by a^{-1} :

$$aa^{-1} \equiv 1 \pmod{m}.$$

- If $a^{-1} = a$, then a is self-invertible.

The unique solution of the linear congruence $ax \equiv b \pmod{m}$, where $(a, m) = 1$, is the least residue of $a^{-1}b \pmod{m}$. ■

Linear Congruences

Modular Inverses

Example: Find $(197)^{-1} \bmod 3000$

STEP 1: FIND THE GCD OF 3000 AND 197

Using Euclid's Algorithm:

$$3000 = 15(197) + 45$$
$$197 = 4(45) + 17$$
$$45 = 2(17) + 11$$
$$17 = 1(11) + 6$$
$$11 = 1(6) + 5$$
$$6 = 1(5) + 1$$

Linear Congruences

Modular Inverses

STEP 2: EXPRESS 1 AS THE DIFFERENCE BETWEEN MULTIPLES OF 3000 AND 197

By backward substitution:

$$
\begin{aligned}
1 &= 6 - 1(5) \\
&= 2(6) - 1(11) \\
&= 2(17) - 3(11) \\
&= 8(17) - 3(45) \\
&= 8(197) - 35(45) \\
&= 533(197) - 35(3000)
\end{aligned}
$$

STEP 3: APPLY MODULO 3000 TO BOTH SIDES

$$1 = 533(197) - 35(3000)$$

$$\Rightarrow 1 \equiv 533(197) - 35(3000) \pmod{3000}$$

$$\Rightarrow 1 \equiv 533(197) \pmod{3000}$$

Answer: 533

Linear Congruences

Solving a Linear Congruence

Steps:

1. Find d = GCD(a, m).
2. Find d mod n → These is the number of solutions available.
3. If $d \mid b$, a solution exists.
4. Divide both sides by d.
5. Multiply both sides by the multiplicative inverse of a.

i. e. $(aa^{-1})\ x = ba^{-1} \pmod{n}$

6. General solution equation is:

$$x_k = x_0 + k\left(\frac{n}{d}\right) \textit{where } k = \{0, 1, 2, \ldots, d-1\}$$

Linear Congruences

Solving a Linear Congruence

Example: Solve 14x ≡ 12 (mod 18)

Solution:

a = 16, b = 12, n = 18

d = GCD(14, 18) = 2

d | b → 12 | 2 = 6 thus solution exists

d mod n = 2 mod 18 = 2 thus 2 solutions

$$14x \equiv 12 \ (mod\ 18)$$

$$\frac{14x}{2} \equiv \frac{12}{2} \left(mod\ \frac{18}{2}\right)$$

$$7x \equiv 6 \ (mod\ 9)$$

$$7 \cdot 7^{-1}x \equiv 6 \cdot 7^{-1} \ (mod\ 9)$$

$$x \equiv 6 \cdot 7^{-1} \ (mod\ 9) \ \rightarrow \ 7^{-1} = 4 \ (Review\ how\ to\ get\ multiplicative\ inverse)$$

$$x \equiv 6 \cdot 4 \ (mod\ 9) = 24 \ (mod\ 9) = 6$$

$$x_0 = 6 \ \rightarrow 1st\ solution$$

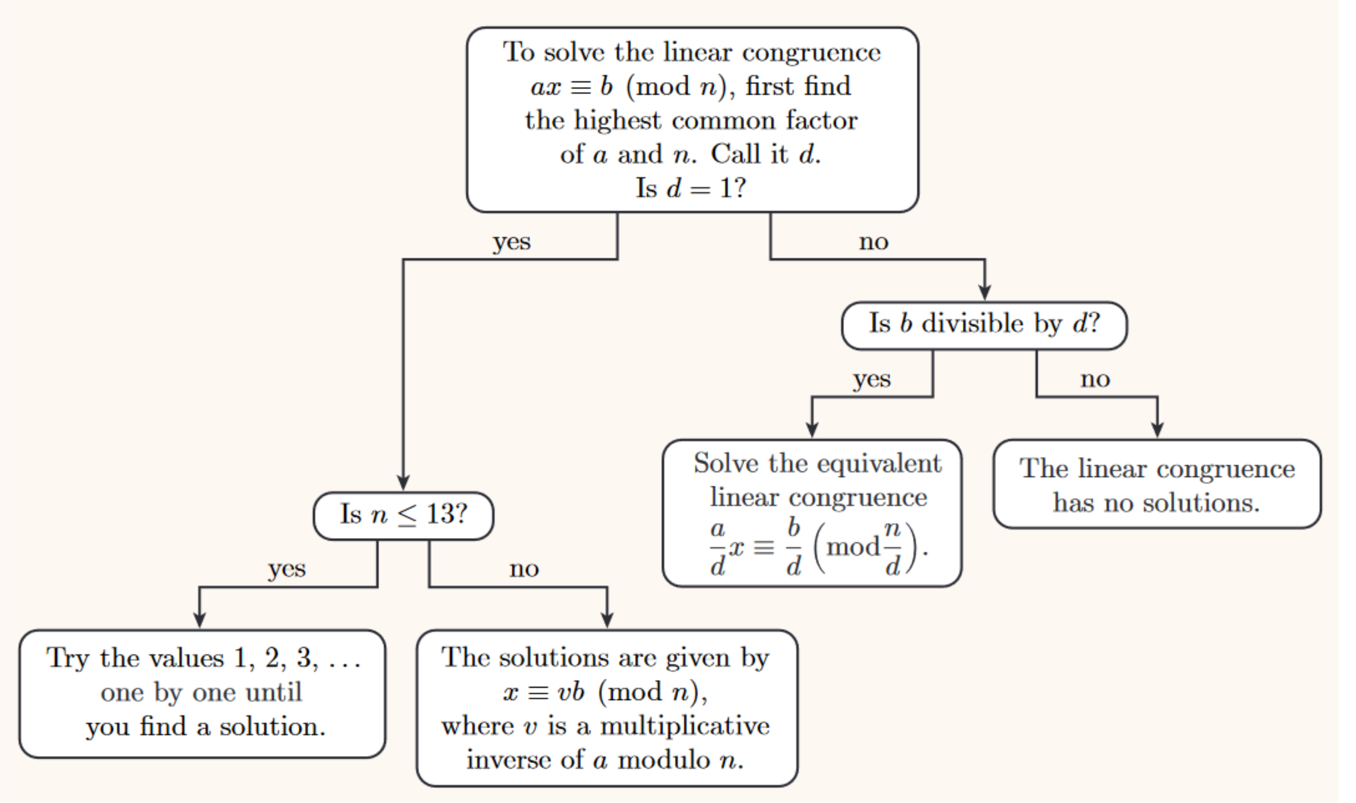
$$x_k = x_0 + k\left(\frac{n}{d}\right) where\ k = \{0, 1, 2, \ldots, 17\}$$

$$x_1 = x_0 + 1\left(\frac{18}{2}\right) = 6 + 9 = 15 \ \rightarrow 2nd\ solution$$

Linear Congruences

Solving a Linear Congruence

A decision tree for solving a linear congruence



Linear Congruences

Activity 2

1. Given the integer 10763732. Test if it is divisible by 2, 3, 4, 5, 9, 7, 11 or 13.
2. Test if a solution exists for $2x \equiv 1 \pmod{3}$ exists. If so, how many are the possible solutions and what are the possible solutions?
3. Solve for $17x \equiv 3 \pmod{29}$

Formative Assessment 9

Check Canvas

SUBMODULE 3 PART 1

Systems of Linear Congruences

OBJECTIVES

- Identify systems of linear congruences.
- Solve a linear congruence using Euclid’s Algorithm.
- Understand the Chinese Remainder Theorem
- Solve systems of linear congruences.

Systems of Linear Congruences

- A set of two or more linear congruences in the same number of variables.

Types of Systems of Linear Congruences

1. Systems in a single variable with pairwise relatively prime moduli.

$$x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}.$$

2. Systems in a single variable with moduli that are not pairwise relatively prime.

$$x \equiv 3 \pmod{6} \qquad x \equiv 5 \pmod{8}$$

3. Systems in two variables with the same modulus (2 x 2 Linear Systems).

$$2x + 3y \equiv 4 \pmod{13} \qquad 3x + 4y \equiv 5 \pmod{13}$$

Systems of Linear Congruences

Systems in a single variable with pairwise relatively prime moduli.

The Chinese Remainder Theorem

The linear system of congruences $\boldsymbol{x \equiv a_i \pmod{m_i}}$, where the moduli are pairwise relatively prime and $1 \le i \le k$, has a unique solution $m_1m_2 \ldots m_k$.

To solve:

b_i (moduli)	$N_i = \frac{N}{N_i}$	x_i (inverse of N_i)	$b_iN_ix_i$

$$\text{Let } N = n_1n_2 \ldots n_k$$

$$x = \sum_{i=1}^{k} b_iN_ix_i \pmod{N}$$

Systems of Linear Congruences

Systems in a single variable with pairwise relatively prime moduli.

The Chinese Remainder Theorem

Example: Solve

$$x \equiv 3 \pmod{5}$$
$$x \equiv 1 \pmod{7}$$
$$x \equiv 6 \pmod{8}$$

Let $b_1 = 5$, $b_2 = 7$ and $b_3 = 8$.
GCD(5, 7) = 1
GCD(5, 8) = 1
GCD(7, 8) = 1
Therefore, the moduli are pairwise relatively prime. Hence, there is a unique solution.

Systems of Linear Congruences

Systems in a single variable with pairwise relatively prime moduli.

The Chinese Remainder Theorem

$Let\ N = 5\ x\ 7\ x\ 8 = 280$

b_i (moduli)	$N_i = \frac{N}{N_i}$	x_i (inverse of N_i)	$b_i N_i x_i$
$b_1 = 3$	$N_1 = n_2 n_3 = 7 \cdot 8 = 56$	$56x_1 \equiv 1 \pmod 5$ $\rightarrow$ inverse is **1**	168
$b_2 = 1$	$N_2 = n_1 n_3 = 5 \cdot 8 = 40$	$40x_2 \equiv 1 \pmod 7$ $\rightarrow$ inverse is **3**	120
$b_3 = 6$	$N_3 = n_1 n_2 = 5 \cdot 7 = 35$	$35x_3 \equiv 1 \pmod 8$ $\rightarrow$ inverse is **3**	630

x = 168 + 120 + 630 = **918**

x ≡ 918 (mod 280) ≡ **78 (mod 280)**

Answer: x = 78

Check:

78 ≡ 3 (mod 5)

78 ≡ 1 (mod 7)

78 ≡ 6 (mod 8)

Systems of Linear Congruences

Systems in a single variable with moduli that are not necessarily prime.

The linear system

$$x \equiv a \pmod{m}$$
$$x \equiv b \pmod{n}$$

has a solution if and only if $(m, n) \mid (a - b)$.

When the linear system is solvable, the solution is unique modulo $[m, n]$.

Hint: Write the first congruence as $x = a + km$ where k is an integer, and then insert this expression for x into the second congruence.

Systems of Linear Congruences

Systems in a single variable with moduli that are not necessarily prime.

Example: Solve: $x \equiv 4 \pmod 6$
$x \equiv 13 \pmod{15}$

Let $a = 4$, $b = 13$, $m = 6$, $n = 16$, $(6, 15) = 3$, $[6, 15] = 30$

Check: $(m, n) \mid (a - b)$? → $(6, 15) \mid (4 - 13)$? → $3 \mid -9$ √ ∴ Solvable

$x = a + km = 4 + 6k$ → $x \equiv 13 \pmod{15}$

$4 + 6k \equiv 13 \pmod{15}$
$6k \equiv 13 - 4 \pmod{15}$
$6k \equiv 9 \pmod{15}$
$2k \equiv 3 \pmod 5$
$k \equiv 4 \pmod 5$

$k = 4 + 5y$; y an integer

$x = 4 + 6(4 + 5y)$
$x = 4 + 24 + 30y$
$x = 28 + 30y$

$x \equiv 28 \pmod{30}$

Systems of Linear Congruences

Systems in two variables with the same modulus.

A 2 x 2 linear system is a system of linear congruences of the form

$$ax + by \equiv e \pmod{m}$$
$$cx + dy \equiv f \pmod{m}$$

has a unique solution if and only if $(\Delta, m) = 1$, where $\Delta \equiv ad - bc \pmod{m}$.

A solution of the linear system is a pair of x and y:

$$x \equiv x_0 \equiv \Delta^{-1}(de - bf) \pmod{m}$$
$$y \equiv y_0 \equiv \Delta^{-1}(af - ce) \pmod{m}$$

where Δ^{-1} is an inverse of Δ modulo m.

Systems of Linear Congruences

Systems in two variables with the same modulus.

Example: Solve:

$$3x + 13y \equiv 8 \pmod{55}$$
$$5x + 21y \equiv 34 \pmod{55}$$

Check: $\Delta \equiv ad - bc \pmod{m} \equiv (3)(21) - (13)(5) \equiv 53 \pmod{55}$.

$(\Delta, m) = (53, 55) = 1$ √ $\therefore$ Solvable

$\Delta^{-1} \equiv 27 \pmod{55}$

$\therefore\ x \equiv x_0 \equiv \Delta^{-1}(de - bf) \pmod{m} \equiv 27\,(21 \text{ x } 8 - 13 \text{ x } 24) \pmod{55} \equiv -7398 \pmod{55} \equiv$ **27 (mod 55)**

$y \equiv y_0 \equiv \Delta^{-1}(af - ce) \pmod{m} \equiv 27\,(3 \text{ x } 34 - 5 \text{ x } 8) \pmod{55} \equiv 1674 \pmod{55} \equiv$ **24 (mod 55)**

Systems of Linear Congruences

Activity 3

1. Solve : $x \equiv 1 \pmod{2}$, $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$
2. Solve : $x \equiv 3 \pmod{6}$, $x \equiv 5 \pmod{8}$
3. Solve : $3x + 4y \equiv 5 \pmod{13}$, $2x + 5y \equiv 7 \pmod{13}$

SUBMODULE 3 PART 2

Special Congruences

OBJECTIVES

- Understand Fermat’s Little Theorem
- Apply Fermat’s Little Theorem to establish congruences with a prime modulus.
- Understand Wilson’s Theorem
- Apply Wilson’s Theorem to establish congruences with a prime modulus.
- Understand Euler Phi Function.

Special Congruences

Fermat's Little Theorem

- A theorem that is used for finding the least residues of powers of integers.

Let p be a prime number, and let a be an integer that is not a multiple of p. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Note: p must not be divisible by a.

Special Congruences

Fermat's Little Theorem

Example: Find the least residue of 4^{20} modulo 7.

By Fermat's little theorem,

$$4^6 \equiv 1 \pmod 7.$$

Therefore

$$4^{12} \equiv (4^6)^2 \equiv 1^2 \equiv 1 \pmod 7,$$

$$4^{18} \equiv (4^6)^3 \equiv 1^3 \equiv 1 \pmod 7,$$

Recall the usual index laws for calculating powers, which tell us that $4^{3\times 6+2} = 4^{3\times 6} \times 4^2 = (4^6)^3 \times 4^2$.

$$\begin{aligned} 4^{20} &\equiv (4^6)^3 \times 4^2 \\ &\equiv 1^3 \times 4^2 \\ &\equiv 16 \\ &\equiv 2 \pmod 7. \end{aligned}$$

So the least residue is 2.

Special Congruences

Wilson'sTheorem

- A theorem that is used for finding remainders involving factorials.
- A theorem that is used for primality tests.

If p is a prime, then $(p-1)! \equiv -1 \pmod{p}$.

Example 1: Primality test

Show that 6 is not prime.

Solution: (6 – 1) ! = 5! = 120 $\equiv$ 0 (mod 6) → not $\equiv$ -1 (mod 6)

Special Congruences

Wilson'sTheorem

Example 2: Finding Remainders

Find the remainder of 97! when divided by 101.

Solution:

First we will apply Wilson's theorem to note that $100! \equiv -1 \pmod{101}$.

When we decompose the factorial, we get that:

$$(100)(99)(98)(97!) \equiv -1 \pmod{101}$$

Now we note that

$100 \equiv -1 \pmod{101}$, $99 \equiv -2 \pmod{101}$, and $98 \equiv -3 \pmod{101}$.

Hence:

$$\begin{aligned}(-1)(-2)(-3)(97!) &\equiv -1 \pmod{101}\\ (-6)(97!) &\equiv -1 \pmod{101}\\ (6)(97!) &\equiv 1 \pmod{101}\end{aligned}$$

Now we want to find a modular inverse of 6 (mod 101).

Using the division algorithm, we get that:

$$\begin{aligned}101 &= 6(16) + 5\\ 6 &= 5(1) + 1\\ 1 &= 6 + 5(-1)\\ 1 &= 6 + [101 + 6(-16)](-1)\\ 1 &= 101(-1) + 6(17)\end{aligned}$$

Hence, 17 can be used as an inverse for 6 (mod 101).

It thus follows that:

$$\begin{aligned}(17)(6)(97!) &\equiv (17)1 \pmod{101}\\ 97! &\equiv 17 \pmod{101}\end{aligned}$$

Hence, 97! has a remainder of 17 when divided by 101.

Special Congruences

Euler's Phi-Function

- Usually used for counting relatively prime numbers.
- Introduced by Leonhard Euler (1707 – 1783).
- This is also called the *totient function* or *indicator function*.
- The symbol ϕ for this function was introduced by Carl Friedrich Gauss (1777 – 1855)

Definition:

For $n \geq 1$, let $\phi(n)$ denote the number of positive integers not exceeding n that are relatively prime to n.

Special Congruences

Euler's Phi-Function

Examples:

$\phi(6) = 2$ →

Number :	**1**	2	3	4	**5**	6
GCD with :		2	3	2	**1**	6

$\phi(20) = 8$

$\phi(19) = 18$

Special Congruences

Euler's Phi-Function

Rule 1: If p is a prime then $\phi(p) = p - 1$.

Example:

$\phi(41) = 41 - 1 =$ **40** GCD(1, 41) = 1 GCD(2, 41) = 1 . . . GCD(40, 41) = 1

Rule 2: If $a = p^n$ is a prime power then $\phi(p^n) = p^n - p^{n-1}$.

Example:

$\phi(32) = 2^5$ so $\phi(32) = 2^5 - 2^4 = 32 - 16 =$ **16**

Special Congruences

Euler's Phi-Function

Rule 3: If GCD(m, n) = 1 then $\phi(mn) = \phi(m)\phi(n)$.

Example:

$$\phi(35) = \phi(7 \times 5) = \phi(7) \times \phi(5) = (7 - 1)(5 - 1) = (6)(5) = \mathbf{24}$$

Another method (Multiplicative Function):

Supposed the only prime divisors of n are $p_1, p_2, \ldots, p_k$.

Then

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\ldots\left(1 - \frac{1}{p_k}\right)$$

Example:

$$\phi(600) = n\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = \mathbf{160}$$

Special Congruences

Activity 4

1. Using Fermat's Little Theorem, find the least residue of 3^{201} (mod 11).
2. Show that 8 is not prime using Wilson's Theorem.
3. Find the remainder of 67! when divided by 71 using Wilson's Theorem.
4. Calculate $\phi(1001)$, $\phi(5040)$ and $\phi(36000)$

Formative Assessment 10

Check Canvas

QUIZ REVIEW

SUMMATIVE ASSESSMENT 3

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